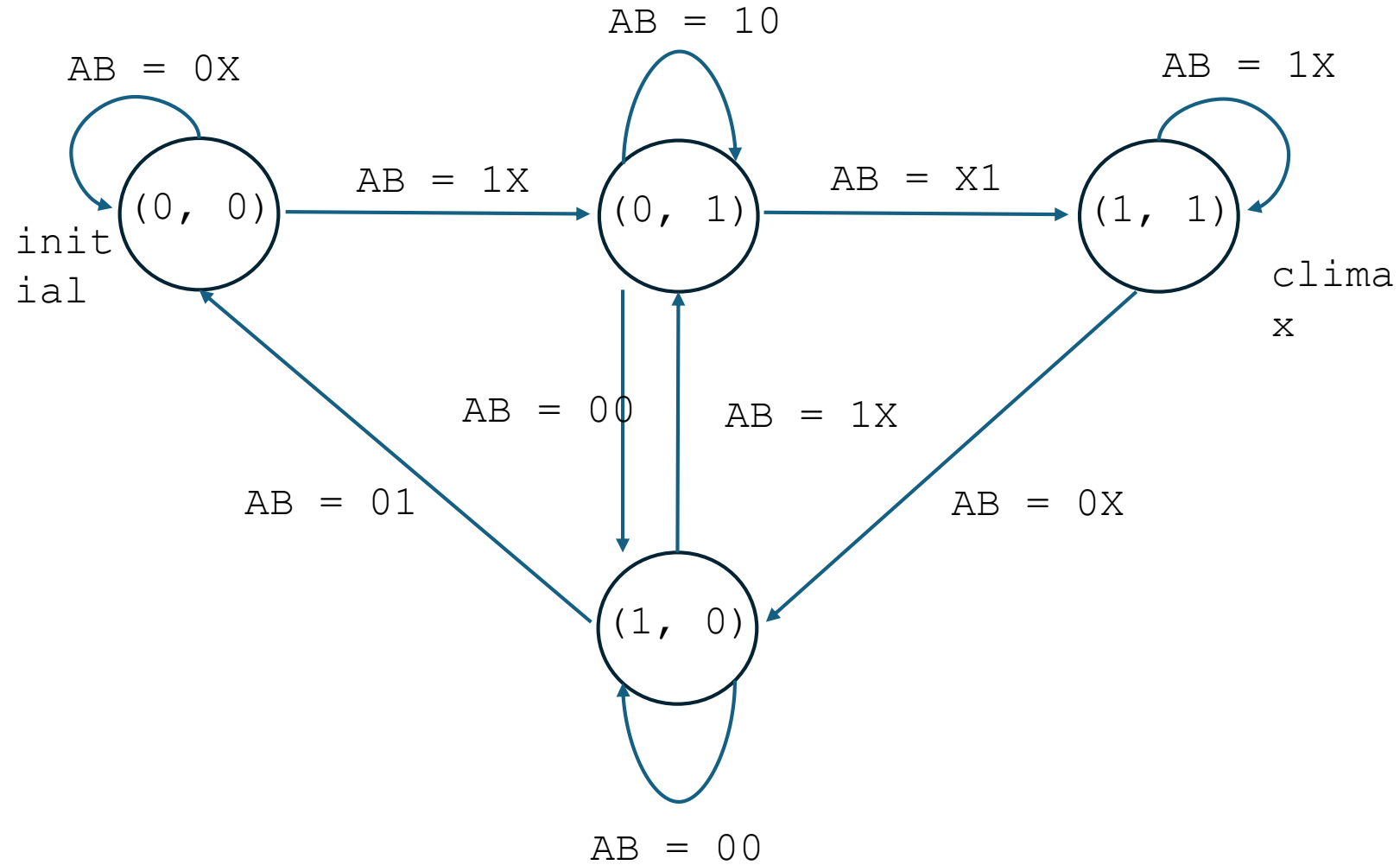


format (L,
R)



Current state (L, R)	Input AB	Next state (L, R)
(0, 0)	1X	(0, 1)
(0, 0)	0X	(0, 0)
(0, 1)	10	(0, 1)
(0, 1)	X1	(1, 1)
(0, 1)	00	(1, 0)
(1, 1)	1X	(1, 1)
(1, 1)	0X	(1, 0)
(1, 0)	00	(1, 0)
(1, 0)	1X	(0, 1)
(1, 0)	01	(0, 0)

4 states num (flip-flop) = $\log_2 4 = 2$

$(0,0) = 00$
 $(0,1) = 01$
 $(1,1) = 11$
 $(1,0) = 10$

F1	F0	input	newF1	newF0
0	0	1X	0	1
0	0	0X	0	0
0	1	10	0	1
0	1	X1	1	1
0	1	00	1	0
1	1	1X	1	1
1	1	0X	1	0
1	0	00	1	0
1	0	1X	0	1
1	0	01	0	0

F1

	AB	A' B	A' B'	AB'
$F_1 F_0$	1	1	1	1
$F_1' F_0$	1	1	1	0
$F_1' F_0'$	0	0	0	0
$F_1 F_0'$	0	0	1	0

$$\text{new}F_1 = F_1 F_0 + F_0 B + F_0 A' B' + F_1 A' B'$$

F0

	AB		A' B		A' B'		AB'	
$F_1 F_0$	1		0		0		1	
$F_1' F_0$	1		1		0		1	
$F_1' F_0'$	1		0		0		1	
$F_1 F_0'$	1		0		0		1	

$$\text{new}F_0 = AB + AB' + F_1' F_0 B$$

output

F1	F0	L	R
0	0	0	0
0	1	0	1
1	1	0	0
1	0	1	0